

Task 4: Aggregate Functions and Grouping — Interview Answers

Objective: Use aggregate functions and GROUP BY to summarize and analyze tabular data.

Key Concepts: Aggregation, Grouping, WHERE vs HAVING, NULL handling, COUNT(DISTINCT), ROUND()

1. What is GROUP BY?

GROUP BY groups rows that have the same values in one or more columns. It is commonly used with aggregate functions such as COUNT(), SUM(), and AVG() to produce summary results for each group.

2. Difference between WHERE and HAVING?

WHERE filters individual rows before grouping or aggregation. HAVING filters groups after GROUP BY and can use aggregate conditions such as HAVING COUNT(*) > 5.

3. How does COUNT(*) differ from COUNT(column)?

COUNT(*) counts every row, including rows containing NULL values. COUNT(column) counts only rows where that particular column is NOT NULL.

4. Can you group by multiple columns?

Yes. For example: GROUP BY department, job_title. This creates groups for each unique department-and-job-title combination.

5. What is ROUND() used for?

ROUND() rounds a numeric value to a specified number of decimal places. Example: ROUND(AVG(salary), 2) rounds the average salary to two decimal places.

6. How do you find the highest salary by department?

Use MAX() with GROUP BY: SELECT department, MAX(salary) AS highest_salary FROM employees GROUP BY department;

7. What is the default behavior of GROUP BY?

GROUP BY returns one summary row for each unique combination of the grouped column(s). Without an aggregate function, it can also be used to return distinct combinations, although SELECT DISTINCT is often clearer for that purpose.

8. Explain AVG and SUM.

AVG() returns the arithmetic mean of non-NULL numeric values. SUM() adds non-NULL numeric values in the selected group or result set.

9. How to count distinct values?

Use COUNT(DISTINCT column). Example: SELECT COUNT(DISTINCT department) FROM employees;

10. What is an aggregate function?

An aggregate function performs a calculation across multiple rows and returns a single summary value per result set or per GROUP BY group. Common examples are COUNT(), SUM(), AVG(), MIN(), and MAX().

Quick SQL Examples

Average salary by department

```
SELECT department, ROUND(AVG(salary), 2) AS average_salary
FROM employees
GROUP BY department;
```

Highest salary by department

```
SELECT department, MAX(salary) AS highest_salary
FROM employees
GROUP BY department;
```

Filter groups using HAVING

```
SELECT department, COUNT(*) AS employee_count
FROM employees
GROUP BY department
HAVING COUNT(*) > 5;
```

Interview Tip

Remember the logical sequence: FROM → WHERE → GROUP BY → HAVING → SELECT → ORDER BY. A simple way to explain it is: WHERE filters rows; GROUP BY creates groups; HAVING filters those groups.